Predicting Downside-Jumps Tech Equity ETFs with Risk-Neutral Skew

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Abstract

In this study, we develop two delta-restricted, moment-based measures of option-implied risk-neutral skew and test their association with next-day downside jumps in technology-sector ETFs. Using the Bakshi–Kapadia–Madan framework on an SVI-parameterized surface, we define a slope signal over the near-at-the-money to 10-delta put wing and a curvature signal spanning the 25-delta put and call regions. Downside events are identified with a daily adaptation of Lee–Mykland, and separate logistic regressions control for at-the-money implied volatility, bid–ask spread, and option volume. Both skew signals have positive, statistically significant coefficients, and each model attains an ROC AUC of 0.91. The results document an association between delta-localized risk-neutral skew and downside-jump labels in the study sample.

1 Introduction

Option prices show how investors price uncertainty across states. In equity markets, out-of-the-money (OTM) puts often carry higher implied volatility than comparable calls or at-the-money options. This negative skew is consistent with demand for downside insurance and with a risk-neutral distribution that assigns relatively more weight to adverse outcomes. Prior evidence links option-skew shape to subsequent market moves, including extreme declines [1]. We test whether localized risk-neutral skew is associated with next-day downside jumps in technology-sector ETF options.

1.1 The Crash-Risk Hypothesis

Our crash-risk hypothesis is that option-implied volatility contains information about the price investors place on downside protection. A steeper put wing may precede or coincide with elevated short-horizon downside risk. This hypothesis concerns prediction, not objective crash probabilities: risk-neutral prices also reflect risk premia, supply and demand, and market frictions.

Technology-sector ETF options provide a useful stress test. The sample spans the COVID-19 shock, the subsequent rally, the 2022 technology-led sell-off, and the recovery through the second quarter of 2024. We examine whether the proposed signals move with statistically identified downside jumps across these sharp market swings. Choosing this period does not imply that technology stocks are uniformly more crash-prone than other sectors. [6, 7]

1.2 From delta buckets to risk-neutral skew

Researchers summarize the implied-volatility surface in several ways. A local slope or finite difference is easy to interpret; a delta-based risk reversal compares the implied volatilities of an OTM put and call at matched absolute deltas. These delta buckets focus on particular regions, such as the two options in a 25-delta risk reversal. Prior work examines demand effects on the surface and the use of delta-segmented smirk measures in return prediction [2, 3]. The comparisons are localized, but draw on only a small part of the option surface.

Fu et al. [10] compare six option-implied volatility measures and find that their predictive content varies by measure and horizon. Their OTM-put-minus-ATM-call measure is especially informative in cross-sectional regressions. Since they define option regions by moneyness rather than delta, their results do not establish that delta thresholds perform best. They motivate our test of localized skew against a distribution-wide moment measure.

Risk-neutral skewness summarizes asymmetry in the distribution implied by option prices. Bakshi, Kapadia, and Madan (2003) derive option-based estimators of risk-neutral moments, including standardized third-moment skewness. Their approach characterizes distributional asym-

metry rather than comparing only two implied volatilities. We represent the implied-volatility surface with SVI, building on Gatheral's work and its arbitrage-free treatment by Gatheral and Jacquier (2014). SVI parameterizes the surface; BKM supplies the option-based moment framework. [9, 5]

Risk-neutral skewness has also been studied outside equity options. Fuertes, Liu, and Tang [11] report excess returns from a commodity-futures strategy sorted on risk-neutral skewness. The returns are larger in contango, and the signal is more stable after controlling for standard commodity factors. Although their setting is different, these results motivate testing the measure in the volatile technology-sector ETF period studied here.

We combine the distributional information in risk-neutral moments with the localization of delta-based measures. The *slope* signal covers the put wing from near ATM through the 10-delta put region. The *curvature* signal spans the 25-delta put and call regions, as does the conventional 25-delta risk reversal. Both are moment-based risk-neutral measures, not raw differences between bucket-average implied volatilities. We test whether this combination adds information beyond an unrestricted moment summary or a conventional delta-only comparison. The analysis estimates each signal separately and does not claim that one outperforms the other.

The empirical analysis covers technology-sector ETF options from Q1 2020 to Q2 2024. We identify downside-jump days with a daily adaptation of the Lee–Mykland procedure, then estimate separate logistic regressions for the two signals. Both specifications control for at-the-money implied volatility, bid–ask spread, and option volume. Section 2 describes the signals, data, event labels, and model; Section 3 reports the estimates; Section 4 discusses their interpretation and limits.

2 Methodology

2.1 Risk-Neutral Skew Signals

We construct two custom measures from the option-implied risk-neutral distribution. Following Bakshi, Kapadia, and Madan [9], we estimate risk-neutral moments, including standardized skewness, from option prices. Before calculating the moments, we represent implied variance across strikes with an SVI parameterization. SVI describes the surface; the BKM framework defines the moment calculation. Our surface specification follows the arbitrage-free treatment of Gatheral and Jacquier [5].

The signals differ from a full-surface risk-neutral skew statistic, which aggregates information across the option-implied distribution, and from a conventional delta measure, which focuses on options near a chosen region. We apply delta windows to the moment-based construction to retain its distributional basis while concentrating on the option regions relevant to each signal. The test asks whether these localized measures add information beyond either benchmark.

The *slope* signal uses the put wing from near ATM to the 10-delta put, capturing changes in downside-tail pricing on the left side of the surface. The *curvature* signal uses the 25-delta put and call regions. Its location matches the conventional risk-reversal comparison, but it is formed from risk-neutral moment information rather than the raw difference between two implied volatilities. We orient both signals so that larger values indicate stronger relative pricing of downside risk, consistent with the positive coefficients reported below.

Let $M_t(D)$ denote the delta-restricted, BKM-based risk-neutral skew statistic at date t for delta region D. The two predictors are

$$S_t = M_t([-0.10, 0)), \tag{1}$$

$$C_t = M_t([-0.25, +0.25]). \tag{2}$$

Thus, the slope signal isolates the put wing near ATM through 10 delta, while curvature spans the 25-delta put and call regions. These are custom moment-based measures, not raw differences

between bucket-average implied volatilities.

2.2 Data

Our options data cover the Invesco QQQ Trust, an ETF tracking the Nasdaq-100 index, from Q1 2020 through Q2 2024. QQQ is technology-heavy, and the sample includes the COVID-19 shock, the 2020–2021 recovery and bull run, and the 2022 technology sell-off. These sharp market swings make the period a useful setting for testing whether implied-volatility skew is associated with downside jumps.

We obtained end-of-day quotes and Greeks for available QQQ options. The data include each contract's strike, expiration, and call or put type, along with implied volatility, trading volume, and open interest. The source provided implied volatilities; we checked them by back-solving option prices with Black–Scholes. The estimates agreed within rounding tolerance. Using the supplied values avoids repeated computation, while the cross-check guards against mismeasurement.

We also use daily QQQ prices and volume. Section 2.3 defines the downside-jump rule; each event date receives a binary label. We pair each day's option features with the label for the following trading day, matching the one-day-ahead question. The rule captures abrupt daily losses but may miss gradual or multi-day declines that never cross the cutoff in one day.

2.3 Jump Detection

We use a daily jump rule inspired by Lee and Mykland [4]. Their nonparametric test was designed for high-frequency returns. Our simpler daily adaptation does not retain those high-frequency properties. It standardizes daily log returns by a rolling estimate of recent dispersion to identify unusually large negative moves.

For daily log returns $r_t = \log(S_t) - \log(S_{t-1})$, we estimate local volatility using the 30-day rolling standard deviation of returns:

$$\hat{\sigma}_t = \mathrm{std}\big(r_{t-29:t}\big) \tag{3}$$

Given this local volatility estimate, we standardize each daily log return as follows:

$$T(t) = \frac{\log(S_t) - \log(S_{t-1})}{\hat{\sigma}_{t-1}} \tag{4}$$

A downside event is flagged when the standardized return falls below the cutoff:

$$T(t) < -3.0 \quad \text{(99.7\% confidence)} \tag{5}$$

The -3.0 cutoff corresponds to a one-sided tail probability of about 0.13% under a standard normal reference distribution. The daily adaptation and the 30-day volatility window determine which observations receive a jump label and matter for interpreting the results.

2.4 Feature Engineering

Alongside the skew measures, the regressions include controls for at-the-money implied volatility, option volume, and bid–ask spread.

ATM Volatility (Level of Volatility). At-the-money implied volatility controls for the overall level of uncertainty. Since volatility itself may rise before a downturn, this control helps distinguish the skew signal from the level of implied volatility.

Trading Volume. We use daily option volume as a control for trading activity. Pan and Poteshman [8] find that excess put buying, measured by the put-call ratio, precedes stock price declines. In our sample, put volume was often elevated before labeled jump days. The model uses a z-score of daily option volume relative to its 20-day average to capture unusual activity.

These controls let us estimate the association between each skew measure and the jump label conditional on the level of implied volatility and trading activity. They do not isolate a causal effect of skew.

2.5 Model

Because the jump label is binary, we estimate separate logistic regressions for slope and curvature. Each model controls for ATM implied volatility, bid–ask spread, and volume. Coefficients are on the log-odds scale; exponentiating them gives odds ratios:

Regression 1 — Slope signal:

$$\Pr(\text{Jump}) = \text{logit}^{-1}\big(\beta_0 + \beta_1 S_t + \beta_2\,\text{ATM IV} + \beta_3\,\text{BidAsk} + \beta_4\,\text{Vol}\big) \tag{6}$$

Regression 2 — Curvature signal:

$$\Pr(\text{Jump}) = \text{logit}^{-1}\big(\beta_0 + \beta_1 C_t + \beta_2\,\text{ATM IV} + \beta_3\,\text{BidAsk} + \beta_4\,\text{Vol}\big) \tag{7}$$

3 Results

Regression 1: Slope signal (S_t)
Pseudo-R^2: 0.333 Log-likelihood: -464.99
Likelihood-ratio test p-value: 2.971×10^{-99}

Variable	Coefficient	Std. Error	z-value	p-value	VIF
Intercept	-0.1255	0.092	-1.371	0.170	1.000
Volume (x_1)	-2.1675	0.184	-11.774	< 0.001	2.072
ATM IV (x_2)	-0.4863	0.131	-3.705	< 0.001	1.595
Bid-Ask Spread (x_3)	0.3706	0.114	3.260	0.001	1.091
Slope signal (x_4)	3.1952	0.212	15.064	< 0.001	2.176

Regression 2: Curvature signal (C_t)
Pseudo-R^2: 0.347 Log-likelihood: -455.53
Likelihood-ratio test p-value: 2.405×10^{-103}

Variable	Coefficient	Std. Error	z-value	p-value	VIF
Intercept	-0.1582	0.094	-1.686	0.092	1.000
Volume (x_1)	-2.2662	0.187	-12.119	< 0.001	2.016
ATM IV (x_2)	-0.4358	0.135	-3.227	0.001	1.634
Bid-Ask Spread (x_3)	0.3866	0.117	3.313	0.001	1.094
Curvature signal (x_4)	3.2256	0.214	15.107	< 0.001	2.181

Table 1: Logistic regressions for next-day downside-jump labels. Coefficients, standard errors, z-statistics, p-values, and variance inflation factors (VIFs) are reported for volume, at-the-money implied volatility, bid–ask spread, and the corresponding risk-neutral skew signal.

Both delta-restricted risk-neutral skew signals are positively associated with next-day downside-jump labels. In the separate logistic regressions in Table 1, the slope coefficient is 3.1952 (standard error 0.212; $z = 15.064$; $p < 0.001$), and the curvature coefficient is 3.2256 (standard error 0.214; $z = 15.107$; $p < 0.001$). Their pseudo-R^2 values are 0.333 and 0.347, respectively, and both models attain an ROC AUC of 0.91. At a false-positive rate near 10%, each ROC curve shows a true-positive rate near 85% in the reported evaluation sample.

The positive estimates mean that higher signal values are associated with greater log odds of the study-defined jump label, conditional on the controls. Slope covers the put wing from near ATM through 10 delta; curvature covers the 25-delta put and call regions. Both retain a defined location on the surface while drawing on risk-neutral moment information. Because the models are estimated separately and no direct test compares the signals, the results do not show that one is superior. They also do not identify whether the association reflects hedging demand, risk premia, or other option-market conditions.

Among the controls, volume is negative and statistically significant in both regressions, bid–ask spread is positive, and ATM implied volatility is negative. The design does not identify a causal explanation for these estimates; they are conditional relationships in this sample. The reported classification results also include 72% recall, a 9% false-positive rate, and an 82% predicted probability for the March 2023 event. These measures depend on the chosen cutoff.

An AUC of 0.91 measures ranking discrimination in this sample, not calibration or economic value. Because the paper does not document a complete time-ordered holdout, neither the AUC nor the threshold-based metrics establish out-of-sample performance.

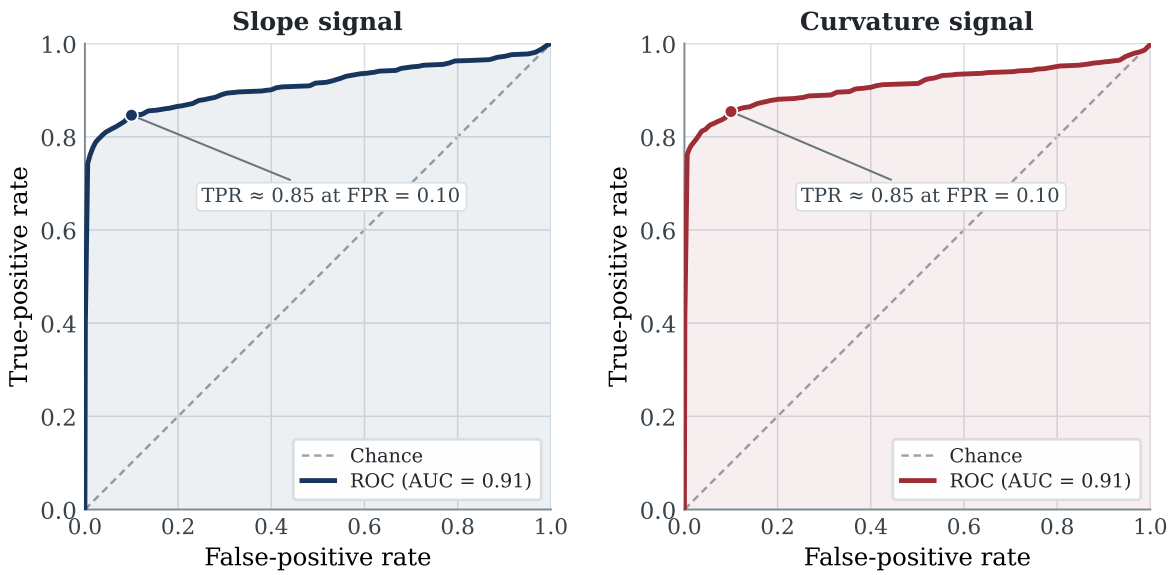


Figure 1: ROC curves for the two models, each with an AUC of 0.91. The curves summarize discrimination in the evaluation sample; see text for interpretation and validation limits.

4 Discussion & Limitations

In QQQ during 2020–2024, both proposed signals are associated with the study-defined downside-jump labels. This is sample-specific evidence, not a causal result or evidence that technology equities are generally more crash-prone. Risk-neutral prices reflect both beliefs about future outcomes and compensation for risk; this study does not separate the two.

The event labels depend on a daily adaptation of Lee–Mykland, whose original test was designed for high-frequency returns. Our daily rule identifies abrupt losses relative to rolling volatility and may miss gradual or multi-day declines that do not cross the threshold in one day. The volatility window and cutoff also determine which observations count as jumps.

The analysis also centers on one ETF and one volatile period. The ROC results measure discrimination in the reported evaluation sample; the study does not establish performance on a fully documented time-ordered holdout, calibration, or trading costs. Tests on other technology-sector ETFs and periods, with a specified out-of-sample design, would show whether these localized moment-based signals generalize beyond QQQ and 2020–2024.

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